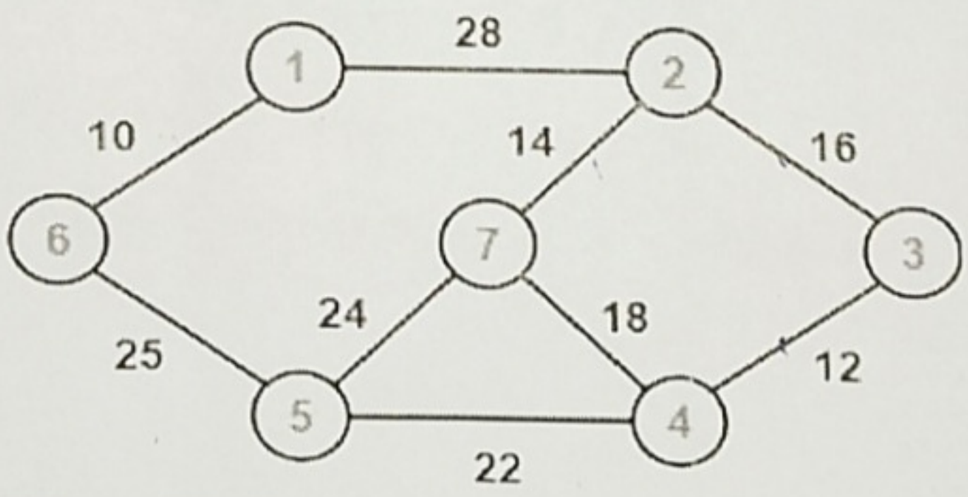
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST-I

Design and Analysis of Algorithms

Class: B. E (CSE)
Semester: IV
Academic Year: 2025-2026

Max Marks: 30
Duration: 1 hr.30min
Date: 24-03-2026

		Marks	BT	CO
	PART A Answer All Questions (Q5 x 2 = 10 M)			
1. a)	Distinguish between best-case and worst-case time complexity.	2	2	1
b)	Write the control abstraction for Divide and Conquer technique?	2	1	2
c)	What are the factors affecting the running time of an algorithm?	2	2	1
d)	State Master's Theorem and derive Time complexity for $T(n) = 4T(n/2) + n^3$	2	3	1
e)	What is Principle of Optimality?	2	1	3
	PART B Answer Any Two Questions (Q2 x 10M = 20 M)			
1.	a) Explain the characteristics of Algorithm and Describe the concept of time–space trade-off by considering any four real time situations.	5	2	1
	b) Solve the recurrent relation: $T(n) = 7T(n/2) + n^2$ Using Substitution method and Masters Theorem	5	3	1
2.	a) Apply Kruskal's algorithm to find minimum spanning tree for the below graph 	4	3	2
	b) Consider the Knapsack instance n = 4, m = 19 $(W_1, W_2, W_3, W_4) = (10, 15, 6, 9)$ $(P_1, P_2, P_3, P_4) = (2, 5, 8, 1)$ Find optimal solution of 0/1 knapsack problem using merge and purge method.	6	3	3
3	a) Compare substitution method, recursion tree method, and Master's Theorem for solving recurrence relations.	6	4	1
	b) Explain job sequencing with deadlines algorithm and also find the solution for the instance n=7, $(P_1, P_2, \ldots P_7) = (3,5,20,18,1,6,30)$ and $(D_1, D_2, \ldots D_7) = (1,3,4,3,2,1,2)$ where P_i is Profit and D_i is Deadline.	4	3	2